

# Lynx: A SmartNIC-driven Accelerator-centric Architecture for Network Servers



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# Background

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- ▶ Modern data centers are increasingly heterogeneous
- ▶ Growing network rates
- ▶ SNICs have been used primarily for accelerating low-level packet processing applications

# Motivation

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- ▶ Diminishing role of CPU in accelerated services
- ▶ Disadvantages of the host CPU-driven design
  - ▶ 가속기 호출 오버헤드
  - ▶ CPU 낭비
  - ▶ Application간의 간섭
- ▶ Limitations of the GPU-centric server design

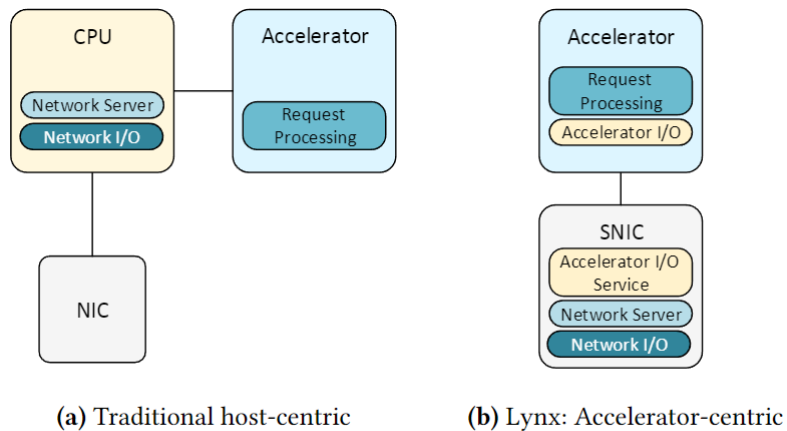
# Benefit

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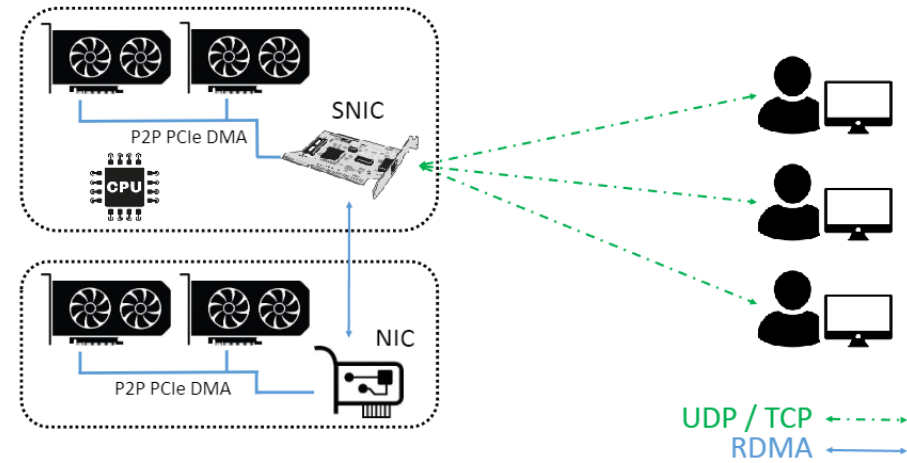
- ▶ Lightweight networking
- ▶ High CPU efficiency
  - ▶ 네트워크 처리와 가속기 제어 부하 감소
- ▶ Performance isolation
  - ▶ SNIC 작업 | 동시 실행 중인 작업
  - ▶ 작업간에 구분이 가능하여 간섭을 없앤다.

# Design

- ▶ Offloading the network server logic to an SNIC
- ▶ Using SNIC RDMA



**Figure 1.** Accelerated network server architectures.



# Main components

- ▶ Network Server
  - ▶ TCP/UDP 처리
  - ▶ Dispatcher, Forwarder가 MQ 전달
- ▶ Message Queues
  - ▶ Rx&Tx Queue
  - ▶ Notification, Completion register(Metadata)
- ▶ Remote Message Queue Manager
  - ▶ RDMA engine

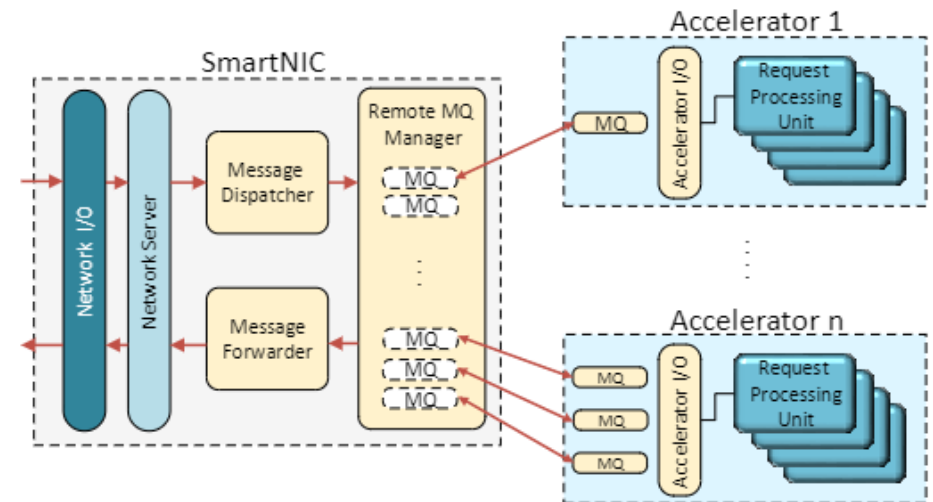
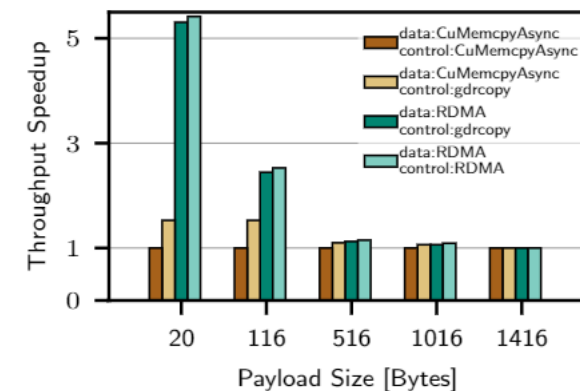


Figure 4. Lynx high level design. MQ: Message Queue. Request Processing Unit represents internal accelerator logic.

# Implementation

- ▶ Using RDMA to access accelerator memory
- ▶ Metadata and data coalescing in mqueues
- ▶ One RC QP per accelerator.



**Figure 5.** Performance of data transfer mechanisms for managing mqueue, relative to cudaMemcpyAsync.

# Evaluation setup

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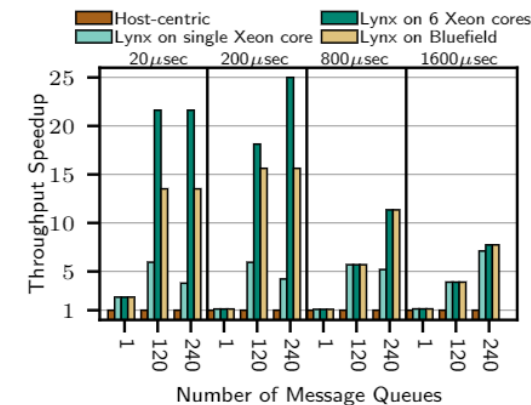
- ▶ 2client 4server
- ▶ Intel Xeon E5-2620 v2 CPU
- ▶ 40Gbps switch
- ▶ 25Gbps Mellanox BlueField SNIC
- ▶ ConnectX-4 Lx EN NICs for remote server
- ▶ 2× NVIDIA K40m, 6× K80 dual GPUs
- ▶ sockperf로 네트워크 성능 측정
- ▶ 실험마다 5회 수행
- ▶ 2초간의 warm up, 20초간 실험 수행



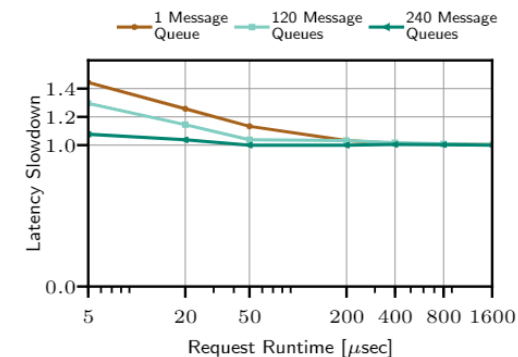
# Evaluation

- ▶ MQ 개수에 따른 성능 측정
- ▶ 실행시간에 따른 성능 측정

- ▶ 지연시간 측정



**Figure 6.** Relative throughput of GPU server implementations for different request execution times (higher is better).



**Figure 7.** Relative latency of a GPU server with Lynx on Bluefield vs. Lynx on 6-core CPU (lower is better). Absolute numbers are in the text.

# Evaluation

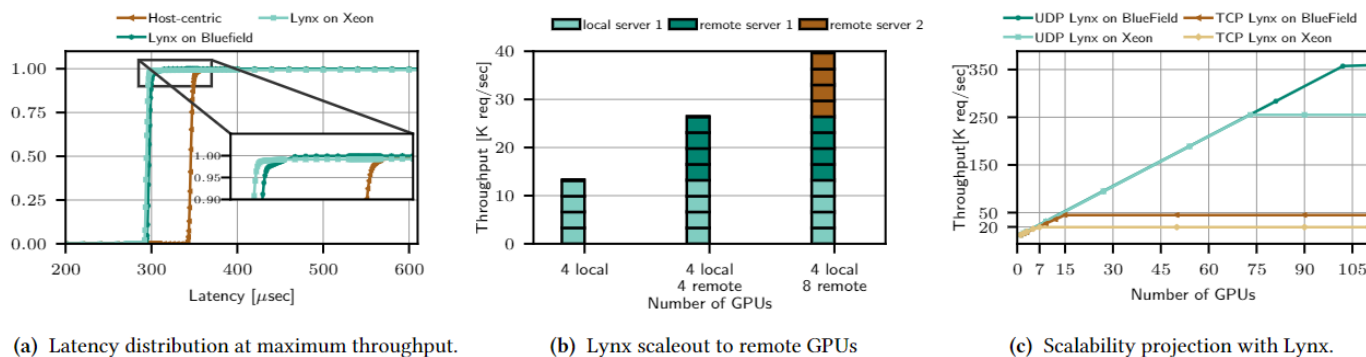


Figure 8. LeNet neural network inference service.

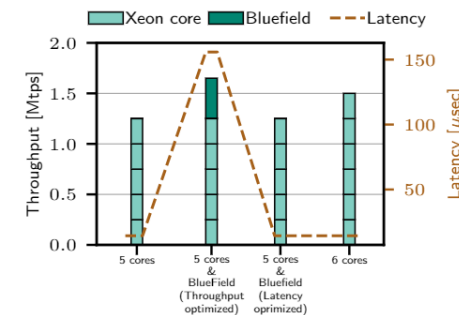


Figure 9. Illustration of the (inefficient) use of Bluefield to run server workloads (memcached) vs. a single Xeon core.

## ▶ GPU 확장성

- ▶ TCP 통신을 사용한 경우 BF 15개 Xeon 7개
- ▶ UDP 통신을 사용한 경우 BF 102개 Xeon 74개
- ▶ 실제 GPU 개수가 아닌 스레드를 사용해 에뮬레이트된 인스턴스 개수
- ▶ BF의 서버 워크로드 처리는 처리량은 증가시키지만 지연시간이 증가

# Conclusion

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- ▶ 전통적 방식에서 Host가 부담하던 네트워크 처리를 SNIC에서 처리하는 방식을 제시
- ▶ 가속컴퓨팅에서 host CPU에 대안을 제시할 수 있음
- ▶ SNIC의 발전으로 더욱 매력적인 대안으로 발전 가능